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CHAPTER

119 OUTLIERS—PART 1: WHAT ARE OUTLIERS?☆

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INTRODUCTION

"What is an 'outlier'?" We will ask this question several times.

It's a happy circumstance (i.e., not exactly a coincidence but not exactly planned, either, that two discussions were written separately and then we noticed how well they fit together) that this new topic follows pretty closely the previous topic, the discussion of Statistics. This is definitely beneficial, if you keep in mind what we told you about statistics while reading what follows; we hope they will be interdependent and shed light on each other. That way you will gain more insight into each topic than reading the explanations for either topic alone, in isolation, will provide.

Why should this be so? Well, the science of statistics is, as a branch of mathematics, very general. It covers descriptions of the way data behave in all other fields of scientific endeavor and even some that are not so scientific (recall one origin of "statistics" as an initial attempt to describe and explain the results of gambling with dice, as we described in Chapter 114). While the subject being investigated in that case was not particularly scientific, applying a rigorous scientific analysis to the situation certainly stood everyone involved in good stead. This is especially so in the case when the underlying subject is itself mathematical, as chemometrics is.

While the scientific field that the original developers of statistics were largely concerned with was astronomy, it is also so with other scientific fields. Statistics, and the results it provides us, can be (and is) applied to chemistry, physics, geology, and virtually every other subject of interest that includes the experimental measurement of one or more physical properties. The measurements in all fields contain error (some of them inherent to the technology, such as we find in techniques such as photon, electron, and particle counting and some due to instrumental or environmental effects). As we have previously

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Our main interest, which is also presumably our readers' as well, is in the application of methods for detection of outliers in situations involving calibration of spectroscopic instruments used for chemical analysis and which also involve application of one or another of the various chemometric algorithms used to perform the analysis. However, we will couch our discussion in a more general terminology because the concepts and applications thereof that we deal with are, in fact, very general. We will, when appropriate, revert to a more specific analysis and applicable terminology when the discussion makes that approach appropriate.

So let's begin at the beginning:

WHAT ARE OUTLIERS?

How can we start? Let's start by trying to figure out just what it is that we're dealing with. The online references given above (mostly) include explicit definitions of the term "outlier." Let's look at some of them:

> A statistical observation that is markedly different in value from the others of the sample (from: http://www.merriam-webster.com/dictionary/outlier).

> "Outliers" are values that "lie outside" the other values (a very literal definition of the term, from: http://www.mathsisfun.com/data/outliers.html).

> An outlier is an observation that lies an abnormal distance from other values in a random sample from a population (from http://www.itl.nist.gov/div898/handbook/prc/section1/prc16.htm which also contains the following caveat: "In a sense, this definition leaves it up to the analyst (or a consensus process) to decide what will be considered abnormal. Before abnormal observations can be singled out, it is necessary to characterize normal observations)."

> An outlier is an observation that lies outside the overall pattern of a distribution (from: http://mathworld.wolfram.com/Outlier.html).

Our hard-copy reference says, "We shall define an outlier in a set of data to be an observation (or subset of observations) which appears to be inconsistent with the remainder of that set of data." (page 4 in Ref. [2]).

The underlying (root) cause of this situation is the basic fact that when physical measurements are subject to random variations due to uncontrollable physical causes, a given measurement can be anywhere within some spread of values, due to the random variations alone. This spread, which can be characterized by the distribution of values within the range of the spread (about which we will have

for one data set with a certain distribution may or may not be an outlier for a set with a different distribution. In this sense, then, we can define an outlier as any data point that cannot be considered to be part of a distribution that characterizes the rest of the data. After all, any finite data set must have a largest value and a smallest value for its observations. Therefore, to claim that a datum is an outlier simply because it is one of the extreme values is clearly misidentifying it as such.

If the suspect datum is not a legitimate part of the data set used for comparison, then it makes sense that a characteristic of that datum is that it will be "far away," or "differ markedly," or "lie outside," or be an "abnormal distance" from the rest of the data, in some way, but the converse is not necessarily true.

It is the dependence on the data distribution that makes the definition (and everything else) about outliers be so fuzzy, as we noted earlier. While all statistics are affected by the randomness inherent in data sets, detection of outliers is perhaps more affected than other aspects of data since almost by definition, an outlier depends on single readings, so it is rare that benefit can be achieved by data processing alone. More important is the attention the data analyst pays to what the data is telling him and not depending on the computer alone to think for him. A nice example was published recently (as of this writing) [4] where, in fact, a data transformation (second derivative) revealed an outlier. But the outlier was observed by "eyeballing" the spectra; the erroneous spectrum was seen to have "wiggles" in a spectral region where the other spectra were smooth. How many data analysts take the time or have the patience to check their data that carefully on a routine basis?

Our intuition about data tends to lead us to an assumption about that data, namely that the distribution is normal (i.e., Gaussian). If that is not the case, however, say if the data comes from, a χ^2 distribution, then many intuitive assumptions about the data will be false. χ^2 distributions tend to be asymmetric with a long tail extending toward higher values. That long tail means that high values (even compared to the other values in the distribution) of χ^2 will be common, and, therefore, ipso facto, not "unusual" or lying at an "abnormal distance" from the rest of the data.

Conversely, since every set of data must have SOME value that is the largest, as well as one that is the smallest value, simply having either of those properties is not enough to flag that datum as an "outlier."

We will continue our discussion of outliers in a future chapter of "Chemometrics in Spectroscopy" coming soon.

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CHAPTER

120 OUTLIERS—PART 2: PITFALLS IN DETECTING OUTLIERS☆

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DETECTING OUTLIERS

Whether we did it well or badly, in Chapter 119 we discussed the concept of what an outlier is. The next point to consider is how to detect an outlier, in the face of the inherent fuzziness of the concept. The fuzz means that we will not always be correct in the assessment of any particular suspect datum, but here we will look at various situations and seek universal methods (which may or may not exist).

A common way to detect outliers is to plot the data and look for readings that are "far away" (in some sense, as judged by the viewer's eye) from the behavior of the rest of the data in the given set under consideration. As we found in Chapter 101, the outlier will visually "pop" out of the rest of the data set in the plot and make themselves apparent. We will discuss those outliers further in Chapter 121. For now, in particular, if the data being examined is the result of a multivariate calibration and the computer is plotting the predicted values from the calibration model versus the reference laboratory results (or even better, the residuals from the calibration versus the reference lab results), we expect the plotted data to cluster around what is called the "45-degree line" (or a flat line, if the residuals are being plotted), with the random error causing the individual data points being normally distributed around that line. This isn't a bad way to examine calibration results, as long as you're aware of the traps and pitfalls, with all the nasty, gooey stuff at the bottom that will eat you up alive, that abound.

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POTENTIAL PITFALLS

For example, as a statistical problem, we must be aware that we have to deal in probabilities. "Everyone knows," for example, that for a normal distribution, 95% of the readings should be within 2 SDs of the center of the distribution and 99% of readings within 3 SDs. Beyond those limits is usually considered the "forbidden zone," in which no samples should be found. So the conclusion jumped to is that a reading beyond 3 SDs is unlikely to occur and therefore any reading beyond 3 SD must be an outlier. Fig. 120-1 shows the normal distribution, as a function of the SD, and we can see how, beyond ±2 SDs, the probability of obtaining a value in that region becomes very small. What is missing in that line of reasoning is the fact that the probability levels given for various values of the normal distribution correspond to the probability of getting that value when selecting a single random sample from the distribution. When more than one sample is taken into consideration, the probability of one or more samples falling in the specified region increases, since any one sample can fall into the "forbidden zone" independently of the other samples. We have discussed this effect previously (see Chapter 1 in Ref. [1]).

In this situation, the actual probability of any of a multiplicity of samples falling into the forbidden zone is given by the equation:

$$P = 1 - (1 - P_i)^n \tag{120-1}$$

where:

P_i is the probability of an individual reading being in the "forbidden zone" (generally 0.05 or 5%)
P is the actual probability that any reading from the set is outside the limit
n is the number of readings (or data points).

For a single sample, then, if a single (random) sample has a probability of 0.95 of being within the "normal" range, then the probability of being outside that range (i.e., in the "forbidden" region) is 0.05.

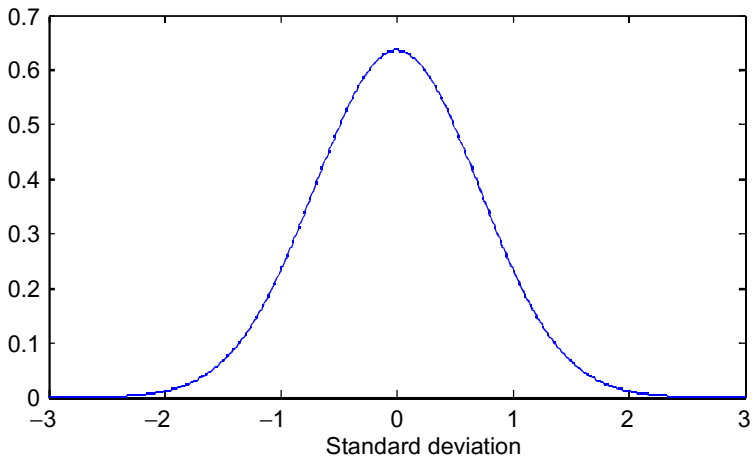


FIG. 120-1

The theoretical (or ideal) normal distribution.

The above-listed outlier tests are all intended to be used for situations where the underlying distribution of values for nondiscordant readings is normal (Gaussian). Barnett [2] also lists another extensive set of tests for discordancy from a normal distribution, for a variety of conditions. That book also presents an extensive set of tests for situations where the underlying distribution is other than normal. We present some of those here also, for reference.

For values that are gamma distributed or exponentially distributed: Barnett provides 16 different tests suitable for various conditions. As with other tests, here we only include one of each type of test that is discussed:

Test statistic for a single upper outlier of an exponential sample:

$$T = \frac{X_n - X_{n-1}}{X_n}$$

For values that are normally distributed, Barnett lists 30 different tests, some of which duplicate ones above, so we forbear repeating them.

For values that are log-normally distributed, take the logarithms of the data and apply a suitable test for data that is normally distributed.

For values that are uniformly distributed, one test uses the following test statistic:

$$T = \frac{X_n - X_{n-2}}{X_n - X_1}$$

Barnett also presents tests for situations where the underlying data distribution is Poisson, binomial, truncated exponential, Pareto, Weibull, and other unusual and exotic distributions, but we forbear to include the tests for all these cases. The alert reader will have noticed that several of the tests described are based on using the same formula for calculating the test statistic. In these cases, the differences between the tests are not in the test statistic itself but in the values corresponding to different critical (threshold) values of the test. This is not very surprising, since it is just those values corresponding to different probabilities that define a given distribution.

Speaking of multiple outliers, as we did briefly, earlier, we have come across a nifty test for multiple outliers in standardization news, the newsletter of the ASTM, now called ASTM International [9], a part of their ongoing promotion of various methods of statistical analysis of data. This test has been given the name Generalized Extreme Studentized Deviate, with the acronym GESD. Without going into all the gory details here, the test was developed to guard against a specific problem: that two outliers, which might otherwise be detected by one of the standard tests, will mask each other by being close together and thereby each will appear less "unusual" than either would alone. The basic test is similar to some of the ones described earlier but is encapsulated in an iterative procedure that also includes modified values of the usual thresholds for the test statistics.

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CHAPTER

OUTLIERS—PART 3: DEALING WITH OUTLIERS☆ 121

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DEALING WITH OUTLIERS

So you've developed a set of data and have a reading that is suspected to be an outlier. You've therefore applied one or more of the tests described in Chapter 120 and confirmed that it is, indeed, an outlier. Now what do you do?

The actions to take can be categorized:

- Delete the discordant reading
- Transform the data
- Accommodate the discordant reading

These are the actions that could be considered the "statistical" approaches to dealing with the outlier. Other approaches can also be considered and, depending on the circumstances, might be preferred. For example, if the situation is that the outlier arises when calibrating a spectrometer for quantitative analysis (using chemometrics, of course, which is the default activity for all our chapters) then the origin of the outlier could be either in the reference laboratory values or among the instrumental values. In either case, an alternative ("nonstatistical") approach would be to identify the source of the discordant value (instrument or lab) by applying knowledge from other sciences and then engage in investigation of the chemistry, physics, background, and so on of the readings to find the fundamental cause. In a pharmaceutical context, for example, this would be what might be required by the FDA and is what would be called "root cause" analysis.

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A variant of this approach is to concentrate all of one's attention on just the "outliers" and ignore the rest of the data, with a view toward learning some new fundamental science, the rationale being that some new and unexpected effect has created the discordancy in the values. An outlier can be an indicator of a scientific accident, of the sort that has led to discoveries such as penicillin, the Big Bang, X-rays, saccharine, Viagra, quinine, smallpox vaccination, Teflon, microwave ovens, LSD, and safety glass [1–3] (don't hold your breath for this, but it DOES happen!). To a lesser degree, we found in Chapter 101 that the presence of a set of outliers that persisted through changes of data transformations and other manipulations of the data was indicative of a (previously unsuspected) systematic effect influencing the data. The fact that the outliers remained unchanged regardless of how the data were treated was a key factor in directing our attention to the reference values as the source of the outliers. The discovery of this effect gave us much information about the meaning of calibration transferability and how to achieve it.

General scientific principles come into play here: Is the effect reproducible? Can the effect be created or avoided at will? Does it lead to predictions of new phenomena? Is there a causal link between the discordant data and fundamental physics, chemistry, etc.? Can you do a controlled experiment (or more generally, what is sometimes loosely called a "statistical experimental design")? Is there a theory (from another science, e.g., chemistry, physics, biology) that can explain the findings? Could you have made a blunder (embarrassing but better to find it out yourself, before someone else does)? Do other scientists get similar results, or has someone at least checked your work? and so on. Here we wish to avoid departing from our mission (such as it is) of describing the chemometric and statistical effects on the data, and so we will not pursue those other topics. However, the reader should keep these alternate considerations in mind; after all, not everything has a statistical explanation! We will now consider the possibilities listed above.

DELETE THE DISCORDANT OBSERVATION

In the context of using chemometric calibration to perform chemical analysis, this is the most common action taken when an outlier is identified. As we saw in the discussion of the nature of outliers in Chapter 119, however, it seems likely that samples summarily rejected this way are sometimes (perhaps even often) misidentified as outliers, when in actuality, they are simply the extreme samples of the distribution. Of course, sometimes the extreme samples are the outliers, if nonlinearity is operative on the samples (e.g., such as we discussed in Chapters 107–111). In that case, however, the ideal fix would be to identify (root cause analysis again, as we described earlier) and correct the cause of the nonlinearity, but that's a topic for a different book.

Alternatively as also described earlier, the distribution could be other than expected. If an outlier is identified based on an expectation of a normal (or at least a symmetric) distribution, but the actual distribution is exponential, or χ^2, or some other asymmetric distribution, then large values are to be expected and summary deletion is obviously an incorrect action.

A consequence of deleting outliers is that the data no longer represents a random sample. Since the discordant observation is, almost by definition, extreme in one characteristic or another, deleting the outliers is a form of censorship of the data. In extreme cases, where multiple "outliers" are deleted from a data set, this is tantamount to fitting the data to the calibration model, rather than a model to the data. If more than two outliers appear to need deletion, then it is more appropriate to reconsider whether they

are actually "outliers" of the data set, or evidence of samples from a different population than the majority of the samples, having been mixed into the data.

A careful statistician will never simply delete observations, even with good reason. Even when some observations must be deleted from a data set, a careful worker will keep a record of the deleted samples: which ones they were, their values, and the rationale for deleting them. This will allow the purported outlier to be added back into the data set, if it turns out that the act of deleting it was erroneous. A set of good recommendations is available in short articles by Tom Fearn [4,5], describing common-sense methods of evaluating suspected outliers and making good decisions about removing them from a data set.

For example, manually entered data is notorious for being rife with data-entry and data-transcription errors. Modern spectroscopic instruments, even low-end ones, generally transfer data to the analysis computer electronically, so this sort of error should not occur. Reference laboratory values, on the other hand, may well be subject to being manually copied, or entered into a database, or transferred to the analysis computer by manual entry, possibly through multiple steps involving manual handling. Typos and other types of errors can occur under these conditions.

TRANSFORM THE DATA

Data transformations are common in spectroscopic calibration practice. This is often done in an ad hoc manner, using the "shot-gun" method of trying every available or conceivable data transform until one of them "works" in some manner. While we can't completely denigrate this approach (if for no other reason than that we've used it ourselves on occasion in the past) we do wish to discourage it and thereby introduce more science into what is currently much more of an "art" of calibration. Therefore when we discuss a technique, or a method, or an algorithm, we do so with the expectation (or at least the hope!) that it will be used as a "rifle": that the new technique will be used only under the specific conditions for which it is suited, to achieve a specific predefined result.

We noted in Chapter 120 that one reason to transform data is to convert the data from having a log-normal distribution to having a normal distribution by taking the logarithm of that data. In chemometric practice, this would typically correspond to the reference laboratory values. Another scenario warranting a transformation of the data is when the error of the reference laboratory measurement is not constant across the range of the analyte (as Gauss prescribed) but increases in proportion to the analyte value, that is, the "relative error" (instead of the absolute error) of the reference method is constant, as happens with some reference laboratory methods. In this case also, the reference values should be replaced with their logarithms, then the errors of those reference logarithm values will be constant. Then, when the calibration calculations are performed, those logarithms are used as the dependent variables, instead of the original reference values.

There is nothing wrong with transforming the reference laboratory values to their logarithms, performing the calibrations for these new (reference laboratory) values and creating a calibration model for those "new" reference lab values. In using the model for analyzing future "unknown" samples, the computation will result in the transformed values being predicted by the model. The user must then remember to back transform the result back to the original domain of the data. When applied to routine analysis, therefore, it is the logarithm of the reference value that is modeled, and the final, post-prediction step needed is to exponentiate the value determined by the calibration model so as to

value (see Fig. 122-6). In both scenarios, however, the cause is the use of an incorrect unit of measure for the reference laboratory readings, and the (proper) cure is, obviously, to use the correct units.

Another "transform" of the data, pertinent to the question of dealing with outliers, is an extension of standard good statistical practice: to collect data from multiple aliquots (at least two, three is preferred and may be required in some cases, as will be described). If two readings are made and one of them is found to be an outlier, then it should be compared with the other reading(s). If they differ by an amount comparable to the magnitude of the outlier, then it seems clear that one of the readings is incorrect, although it may not be obvious which one that is. If data from three or more replicates were collected, however, then a "vote" can be taken; the two that agree are most likely correct, and the one that disagrees, especially if it is the discordant reading, is probably wrong. It may not be clear why that reading is different, but this may be a case where the type of "root cause analysis," as described earlier, might be warranted. If none of the three (or more) readings is noticeably different from the rest, then it might be appropriate to average all the like readings together, prior to performing the calibration calculations.

The data may also be "windsorized" by averaging together the highest and lowest values with their respective nearest neighbors, as a way to reduce the influence of a possible outlier. This may be ineffective if the data is multivariate, since the outlier may not correspond to any individual extreme reading.

ACCOMMODATE THE DISCORDANT OBSERVATION(S)

"Accommodation" of an outlier is not necessarily a way to identify it. It is often a technique for reducing or eliminating the effect of a possible outlier on the results of the calculations, whether any outliers are present or not.

There are many methods of accommodating outliers. One that is rarely used but probably should be used more commonly is what is called "robust" methods of analysis. These robust methods replace our usual methods of calculation with methods that produce essentially the same results but are resistant to the effects of outliers. The simplest example of a robust method is the use of the median, instead of the mean, as the measure of the central value of a set of data. Since the value of the median depends almost entirely on the numbers at or near the center of the data, the values at the extremes are virtually immaterial. The largest and smallest values can be correct or off by a large factor, but that difference has no influence at all on the value of the median. Contrast that with the mean, where the influence of the extreme values increases as the square of the difference between the mean and the values themselves, so that the incorrect extreme value has undue influence on the results. There are robust equivalents to most of the common chemometric methods that are popular, including MLR and PCR. On the other hand, making that decision is "all or nothing" since the robust statistics have different properties than the parametric statistics that we are used to working with (e.g., as we showed earlier in Chapter 116, the mean is a least square estimator, whereas the median is not).

A somewhat less extreme way of accommodating (potential) outliers is to use *weighted* calculations. This permits the use of the standard algorithms and methods of analysis. The key here is the fact that outliers are, in general, the extreme observations of some characteristic of the data. This being the case, if the weighted versions of the standard algorithms are used; with the weights corresponding to the degree of extremeness of the pertinent characteristic, the more extreme the value of the characteristic, the lower the weight assigned to the sample containing that extreme value and the less its influence on the final results. This approach has the advantages of retaining and using all the data, while still

minimizing the effect of the outlier on the results. Equations for weighted as well as unweighted calibration algorithms for implementing MLR, PCR, and PLS algorithms are described in ASTM standard E1655-05 [7].

SUMMARY

As noted at the beginning of this chapter, deletion of an apparent outlier is the most common way that most spectroscopists know to deal with them. As we have seen, however, there are alternate methods available to take outlier readings into account, in spectroscopic analysis. Add that to the fact that in much calibration work, a great excess of enthusiasm is shown in the deletion of purported or suspected outliers, so we hope that having alternate methods available to deal with them can reduce this overenthusiasm and preserve data sets.

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